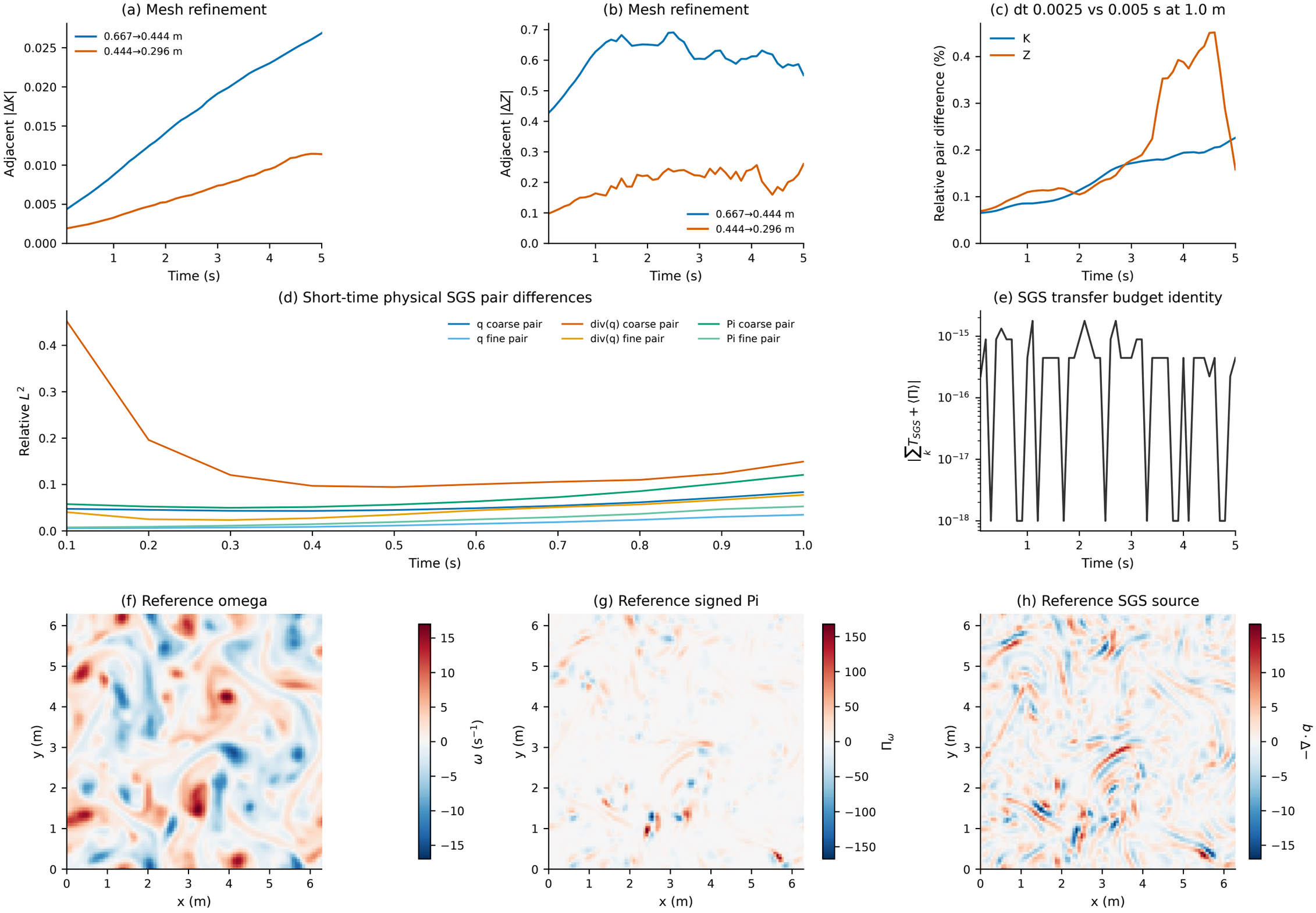

(a) Mesh refinement
0.667→0.444 m
0.444→0.296 m
Adjacent |ΔK|
Time (s)
(b) Mesh refinement
Adjacent |ΔZ|
0.667→0.444 m
0.444→0.296 m
Time (s)
(c) dt 0.0025 vs 0.005 s at 1.0 m
K
Z
Relative pair difference (%)
Time (s)
(d) Short-time physical SGS pair differences
q coarse pair
q fine pair
div(q) coarse pair
div(q) fine pair
Pi coarse pair
Pi fine pair
Relative L^2
Time (s)
(e) SGS transfer budget identity
$|\sum_k T_{SGS} + \langle\Pi\rangle|$
Time (s)
(f) Reference omega
y (m)
x (m)
ω (s^{-1})
(g) Reference signed Pi
y (m)
x (m)
Π_ω
(h) Reference SGS source
y (m)
x (m)
$-b \cdot \Delta$